

SHUMEN, Bulgaria

WMO: 155440

Lat: 43.267N

Lon: 26.933E

Elev: 722

StdP: 14.32

Time Zone: 2.00 (EUE)

Period: 82-90

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	15.9	4.7	7.2	12.5	8.7	8.8	17.8	35.4	45.4	30.7	37.1	3.0	N/A	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	88.7	68.3	85.6	67.1	82.7	65.7	70.5	84.2	68.7	81.3	67.4	79.2	6.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	99.7	75.9	64.5	93.2	73.2	62.9	88.3	72.1	34.9	84.0	33.2	81.2	32.2	79.5	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	22.1	18.3	DB	5.1	94.7	5.8	4.2	1.0	97.8	-2.4	100.2	-5.7	102.6	-9.9	105.7	(4)
(5)			WB	4.5	75.0	5.7	3.7	0.4	77.6	-2.9	79.8	-6.1	81.9	-10.2	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	33.0	33.6	41.1	51.8	60.5	66.3	70.3	69.9	64.6	52.7	40.8	36.1	(6)	
	DBStd	15.56	8.25	9.74	9.97	6.67	6.15	4.60	4.97	4.60	5.26	7.18	8.21	8.20	(7)	
	HDD50	2134	528	463	307	56	3	0	0	0	0	52	291	434	(8)	
	HDD65	5311	992	879	741	396	164	39	8	13	70	387	727	895	(9)	
	CDD50	2802	1	4	30	112	329	488	629	618	437	135	14	5	(10)	
	CDD65	505	0	0	0	2	25	77	172	166	57	6	0	0	(11)	
	CDH74	4530	0	0	0	46	247	642	1543	1447	552	53	0	0	(12)	
	CDH80	1296	0	0	0	9	30	134	537	444	128	14	0	0	(13)	
(14) Wind	WSAvg	5.2	5.2	5.8	5.8	6.1	5.1	6.0	5.1	4.7	4.0	4.3	4.6	5.4	(14)	
(15) Precipitation	PrecAvg	22.0	1.4	1.5	1.5	2.0	2.6	3.0	2.0	1.7	1.4	1.5	1.9	1.9	(15)	
	PrecMax	35.5	3.8	4.1	4.3	3.7	6.4	6.5	4.3	5.1	4.1	4.7	6.9	8.7	(16)	
	PrecMin	13.8	0.1	0.4	0.2	0.4	0.2	0.8	0.4	0.0	0.0	0.1	0.2	0.1	(17)	
	PrecStd	4.5	1.0	0.9	1.1	0.8	1.7	1.4	1.2	1.1	1.1	1.1	1.3	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	60.3	72.1	79.5	83.5	89.0	96.5	90.8	88.3	80.4	64.5	62.8	(19)	
		MCWB	46.8	49.5	53.4	57.9	64.3	70.3	73.7	68.1	64.1	63.6	54.6	50.8	(20)	
	2%	DB	52.7	56.1	66.2	73.1	79.4	83.3	90.1	87.5	83.5	72.7	60.3	56.1	(21)	
		MCWB	45.1	47.1	51.3	56.2	61.8	66.9	71.1	67.1	63.7	58.7	53.1	48.3	(22)	
	5%	DB	48.5	51.7	61.7	69.2	76.7	80.5	85.6	85.0	80.0	68.4	56.6	51.7	(23)	
		MCWB	43.1	45.0	49.7	54.6	60.6	65.9	68.4	66.3	63.1	57.4	50.6	45.3	(24)	
	10%	DB	44.3	46.8	57.4	65.2	73.6	77.7	82.1	82.2	76.7	64.5	52.7	47.9	(25)	
		MCWB	40.0	41.6	47.5	52.7	59.5	64.4	66.5	65.3	62.0	55.4	48.5	43.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	50.6	55.0	60.3	66.5	71.0	75.5	70.4	68.0	63.9	55.8	51.7	(27)	
		MCDB	54.5	57.4	67.0	74.3	80.0	84.4	91.9	85.0	76.8	79.0	62.9	58.7	(28)	
	2%	WB	45.7	48.3	52.8	57.8	64.1	68.7	72.3	68.9	66.2	60.3	53.7	49.3	(29)	
		MCDB	51.7	55.8	64.2	70.7	75.5	81.3	87.0	82.9	78.7	68.8	59.9	54.7	(30)	
	5%	WB	43.2	45.4	50.3	55.3	62.4	67.3	69.7	67.8	64.8	58.3	51.3	46.2	(31)	
		MCDB	48.5	50.5	61.1	67.4	73.5	78.2	81.9	81.1	76.4	66.5	55.1	51.5	(32)	
	10%	WB	40.3	42.2	47.7	53.3	60.6	65.7	67.7	66.7	63.2	56.2	49.0	43.1	(33)	
		MCDB	43.7	46.7	56.6	64.2	70.5	75.3	79.3	78.9	73.6	63.4	52.6	47.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.8	15.0	18.3	20.0	18.6	19.8	20.7	21.3	17.3	13.4	11.7	(35)	
		MCDBR	19.2	20.2	24.4	27.2	24.8	24.0	26.1	25.8	27.0	24.4	19.8	19.7	(36)	
	5% WB	MCWBR	13.5	13.0	12.7	13.0	10.6	10.4	9.8	9.5	10.5	11.6	12.6	12.3	(37)	
		MCWBR	18.8	17.6	23.7	24.8	23.3	22.1	23.0	23.2	23.5	21.7	17.2	17.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.364	0.401	0.449	0.425	0.429	0.438	0.448	0.421	0.385	0.358	0.336	(40)	
		taud	2.334	2.262	2.201	2.075	2.218	2.239	2.227	2.194	2.242	2.328	2.352	2.344	(41)	
	Ebn at Noon		248	259	263	259	270	268	264	257	254	249	237	235	(42)	
		Edh at Noon	26	34	40	49	44	43	43	43	38	31	26	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		474	709	1064	1485	1856	2034	2069	1859	1353	868	522	403	(44)	
	RadStd		65	93	99	164	138	153	139	98	112	90	68	51	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	-194	+146	+90	

Nomenclature: See separate page